TECHNICAL ARCHITECTURE // MEMORY SYSTEMS

# When Memory Lies: The Hidden Complexity of DDR Interfaces

A bottom-up journey from a single leaking capacitor to billions of transfers per second.

# About me

ENGINEER PROFILE // ALÉN SPACE

## Reyders Morales

Embedded Engineer at Alén Space

TECHNICAL FOCUS AREAS

- Software Defined Radios (SDRs)
- RTEMS, FreeRTOS & Linux
- Embedded Electronics & Hardware
- Satellite In-Orbit Support

CORE RESPONSIBILITIES

**OS & Hardware Level Support**
Operating system support at hardware level for SDR payload lines.

**Electronics & Board Bring-Up**
Electronics bring-up, testing, and on-board hardware integration.

**Driver Development**
Device driver development for RTEMS, FreeRTOS, and Linux platforms.

**Customer Application Integration**
Customer support and technical assistance for custom application integration.

**In-Orbit Payload Operations**
Ongoing technical support and maintenance for active in-orbit payloads.

# Alén Space: Brief History

## 2008

The University of Vigo's **Strategic Aerospace Group** was founded by the future team of Alén Space.

## 2012

Launch of the first Spanish nanosatellite: **Xatcobeo**.

## 2019

**Alén Space** is officially established as an independent company.

## 2023

Surpasses **50 professionals** and expands facility work areas.

Acquisition agreement by the **GMV Group**.

## 2026

Surpasses **110 professionals** and has more than **17 satellites** in orbit.

# What is Embedded? What is an SDR?

CONCEPT 01 // DEDICATED HARDWARE & SOFTWARE

## Embedded Systems

A specialized computing system combining tailored hardware and dedicated firmware built to perform a single specific function with high reliability.




**KEY CHARACTERISTICS:**

- **Purpose-Built:** Designed for precise task execution (e.g., satellite control).
- **Resource Constrained:** Optimized for low power, memory, and footprint.
- **Real-Time Operation:** Deterministic response time for critical tasks.

CONCEPT 02 // SOFTWARE DEFINED RF

## Software Defined Radio (SDR)

A radio communication system where traditional hardware components (mixers, filters, modulators) are replaced by flexible software algorithms.




**KEY CHARACTERISTICS:**

- **Reconfigurable:** Change protocols in-orbit via simple software updates.
- **Multi-Band:** Process wide radio frequencies without hardware changes.
- **Space Efficient:** Reduces physical payload footprint on satellites.

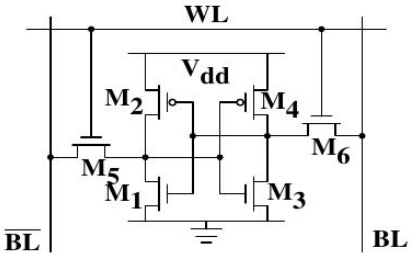
# RAM Types: SRAM vs DRAM

CONCEPT 01 // BISTABLE LATCH (6T CELL)

## SRAM (Static RAM)

Retains data as long as power is supplied without requiring refresh cycles. Uses a 6-transistor (6T) flip-flop structure per bit.

- **Speed:** Ultra-fast access (sub-nanosecond latency)
- **Use Case:** CPU Caches (L1/L2/L3) & registers
- **Trade-off:** Low density, expensive, higher active power

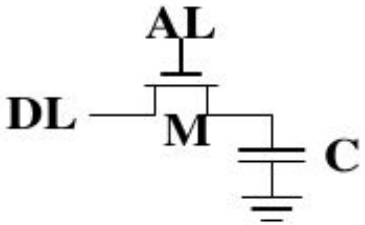



CONCEPT 02 // CAPACITOR CHARGE (1T1C CELL)

## DRAM (Dynamic RAM)

Stores bits as electrical charge in capacitors (1T1C). Inherently leaky, requiring constant refresh cycles to retain state.

- **Refresh Interval:** Must refresh every ~64ms
- **Use Case:** Main System Memory (DDR4 / DDR5)
- **Trade-off:** High density & low cost/bit, higher latency



# DRAM Architecture: Basic Memory Array Structure

## ARRAY ORGANIZATION & ADDRESSING

**1. 1T1C Memory Cells:** Each bit consists of a single access transistor and storage capacitor at grid intersections.

**2. Row Address Selection (Wordlines):** Selects a row via address inputs ($a_0$, $a_1$), enabling all access transistors along that line.

**3. Column Address Selection (Bitlines):** Multiplexes bitlines via address inputs ($a_2$, $a_3$) to route target cell charges to I/O Data lines.

**4. Multiplexed Addressing:** Splitting addresses into row and column cycles significantly reduces required chip pin counts.

### 2D GRID SCALABILITY

An N × M matrix connects N · M memory cells using only N + M control lines, maximizing packing density.

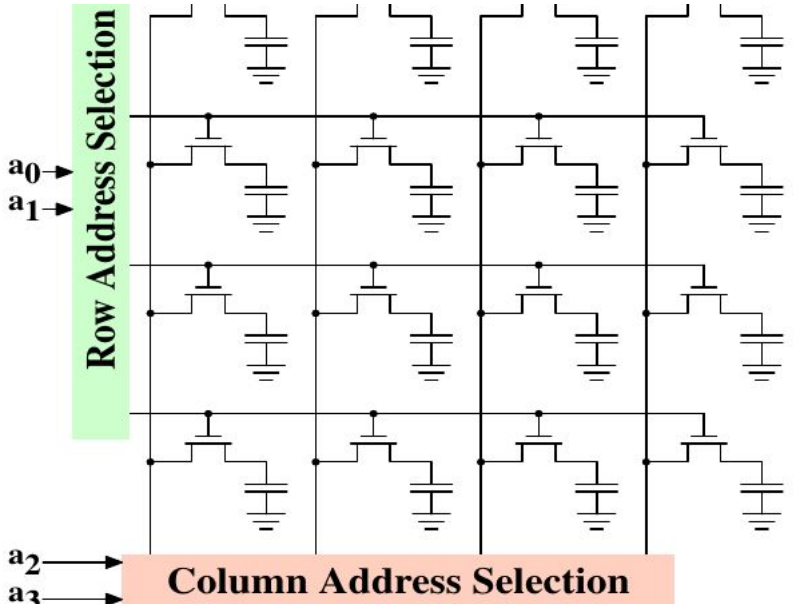



FIG – 4x4 DRAM MATRIX & ADDRESS SELECTION SCHEMATIC

# DRAM Read Timing: Understanding tRCD & CAS Latency

## TIMING PARAMETERS & LATENCY

**1. Row Activation (RAS#):** Asserting RAS# latches the Row Address and opens the row into the sense amplifiers.

**2. tRCD Delay (RAS-to-CAS Delay):** Required physical wait time between activating a row and issuing a Column Read command.

**3. Column Selection (CAS#) & CL:** CAS# selects the column. **CAS Latency (CL)** is the delay before valid data hits the bus.

**4. Data Burst (DQ Output):** Data is output on the DQ lines in synchronous bursts aligned with the clock signal.

**tRCD + CL**

**READ ACCESS LATENCY**

Total timing bottleneck required to read the first word from a closed row.

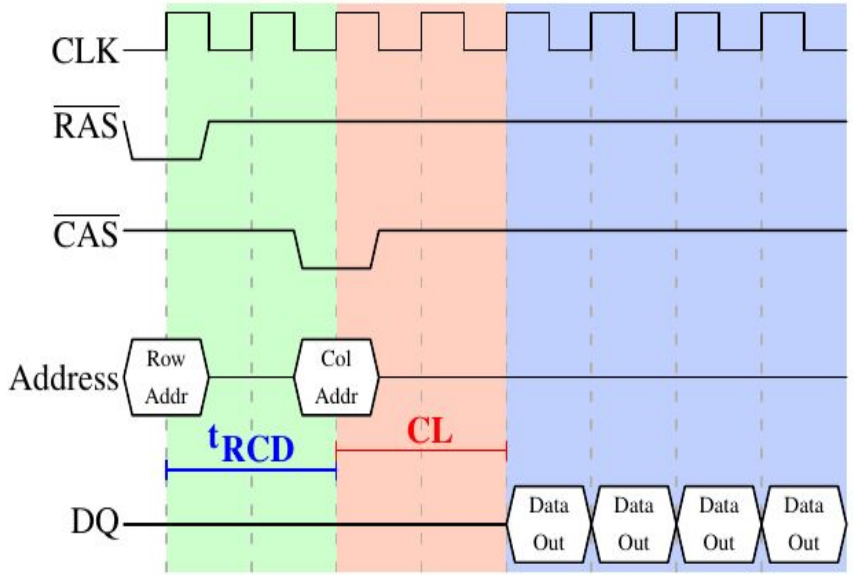



FIG 1.2 – DRAM READ COMMAND & TIMING WAVEFORM

# DRAM Operations: Row Precharge & Activation

## ROW PRECHARGE & ACTIVATION CYCLE

**1. Row Precharge (tRP):** Closing an active row and restoring bitlines to VREF before another row can be accessed.

**2. Row Activation (RAS#):** Asserting RAS# latches the row address and opens wordlines into the sense amplifiers.

**3. tRCD Delay:** Required physical wait time between activating a row and issuing a Read/Write command.

**4. Bank Switching Overhead:** Accessing a different row within the same bank forces a sequential Precharge-then-Activate delay.

**tRP + tRCD**

**ROW SWITCHING OVERHEAD**

Total latency penalty incurred when closing an active row and opening a new one.

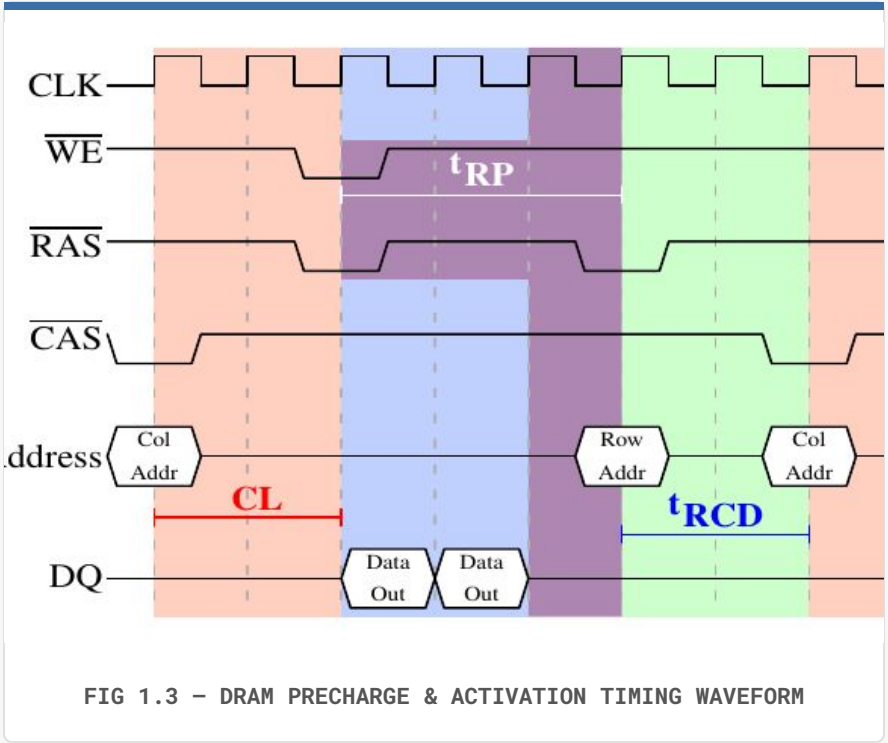



FIG 1.3 – DRAM PRECHARGE & ACTIVATION TIMING WAVEFORM

# What an embedded engineer must understand during DDR bring-up:



STEP 01 // HARDWARE

## Power & Reset

Verify power rail sequencing, stable VREF, clean reference clocks, and proper timing on the RESET# de-assertion.

STEP 02 // JEDEC

## Mode Registers

Program Mode Registers (MRs) with exact timing values (CAS latency, tRCD, tRP, burst length) matching DRAM chip specs.

STEP 03 // PHY

## DDR Training

Execute Write/Read Leveling, ZQ calibration, and VREF data-eye tuning to compensate for PCB trace length skews.

STEP 04 // INTEGRITY

## Stress Testing

Validate stability under high load (memtester, stressapptest) across voltage/temperature boundaries to reveal margin issues.

# Why is it so complex to bring up DDR in custom hardware?



CHALLENGE 01 // ANALOG

## Signal Integrity

Gigahertz clock rates require sub-millimeter trace length matching, tight impedance control, and minimal crosstalk across high-density lines.

CHALLENGE 02 // PHY

## DDR Training

The PHY layer must execute complex dynamic routines (Write/Read leveling, VREF eye centering) to compensate for board skews.

CHALLENGE 03 // TIMING

## JEDEC Protocols

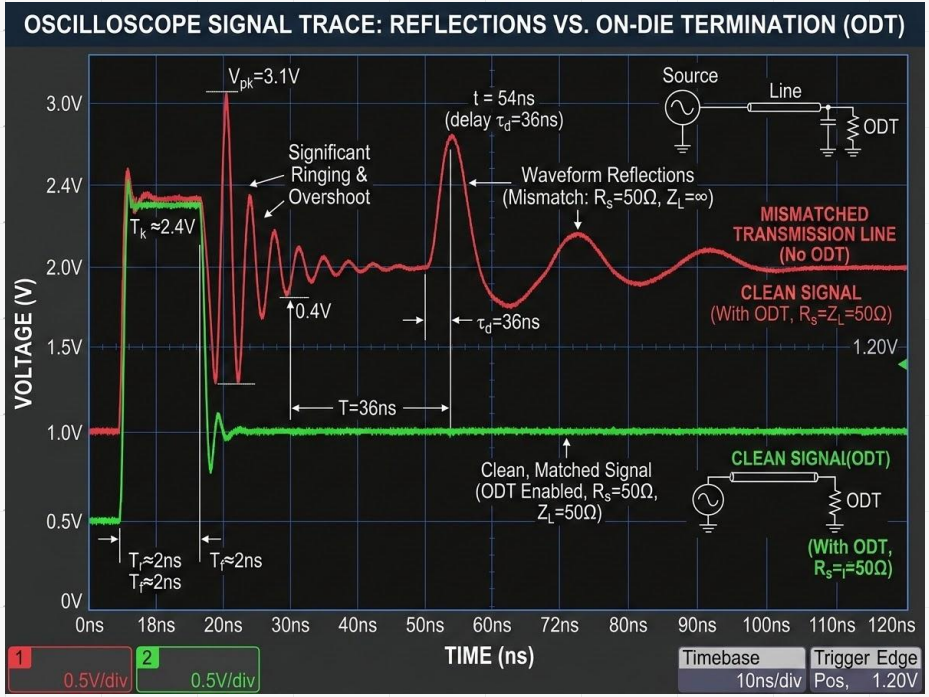
Dozens of interdependent timing parameters (tRP, tRCD, tRAS, CL) must be configured in controller registers with picosecond accuracy.

CHALLENGE 04 // POWER

## Power Delivery

Simultaneous switching noise (SSN) and power rail ripple can shrink timing margins, requiring clean reference voltages and proper decoupling.

# High-speed DDR requires precise impedance matching to prevent signal reflections and crosstalk.




## ON-DIE TERMINATION (ODT)

At gigatransfer speeds, memory traces act as transmission lines where mismatches distort waveforms.

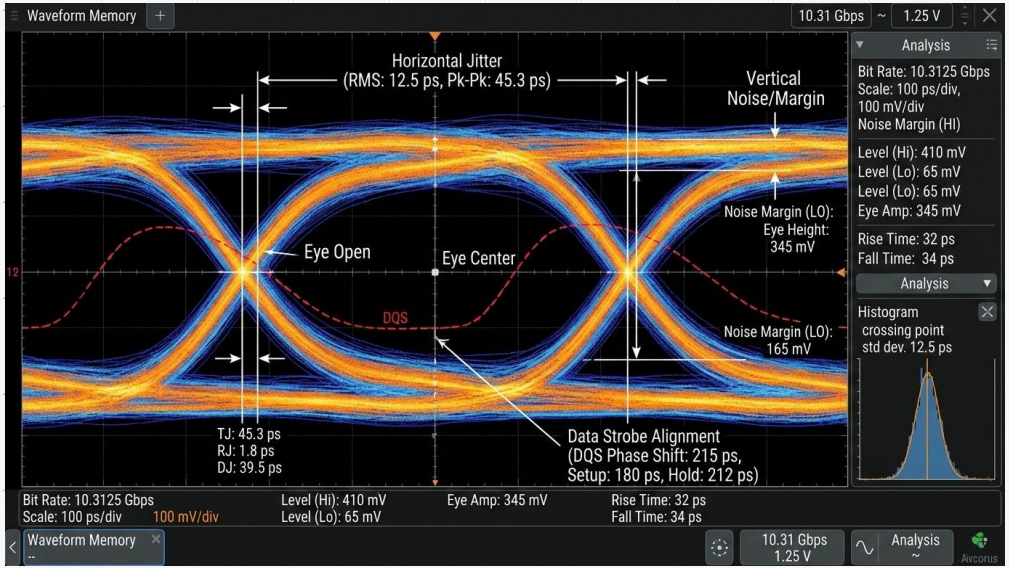
**ODT dynamically applies resistors** to absorb signal reflections and maintain clean waveform traces.

## CROSSTALK & SIGNAL INTEGRITY

Dense trace routing introduces electromagnetic crosstalk across parallel memory channels.

Engineers enforce **differential signaling** for clock lines and strict **length-matching rules** across the PCB layout.

# Source-synchronous clocking captures data within shrinking valid signal windows.




## TIMING & SIGNAL INTEGRITY

DDR transmits data on both the rising and falling edges of the clock.

Instead of relying on a global clock, DDR uses a **Data Strobe (DQS)** sent alongside the data (DQ) to minimize timing skew.

Engineers analyze Eye Diagrams to visualize signal quality; a wider 'eye' indicates a larger valid window for capturing bits before voltage or timing margins collapse.

# DDR evolution relies on increased prefetch buffers and lower voltages to scale bandwidth.

**SDR DRAM**

## 2n / 3.3V

Legacy baseline operating parameters

**DDR3 / DDR4**

## 8n / 1.2V

Intermediate prefetch scaling & drop

**DDR5 ERA**

## 16n / 1.1V

16 data words per internal clock cycle

From SDR to DDR5, core DRAM array speeds have barely increased. Instead, bandwidth scales via the prefetch architecture. DDR5 uses a 16n prefetch, pulling 16 data words per internal clock cycle to feed the high-speed external bus.

Operating voltages have simultaneously dropped from 3.3V in early SDR to 1.1V in DDR5 to manage thermal limits at multi-gigahertz frequencies.

# HARDWARE ARCHITECTURE / MEMORY SYSTEMS

> *"Reliable computing is an illusion actively maintained by complex memory controllers."*
>
> CORE SYSTEM ARCHITECTURE PARADIGM

## 01 / PHYSICAL HARDWARE LIMITATIONS

DDR memory demonstrates that fundamental hardware is inherently unreliable, leaky, and noisy. Billions of transfers per second are only possible through layers of physical mitigation, dynamic calibration, and error correction.

## 02 / SOFTWARE & PERFORMANCE IMPACT

Understanding this bottom-up complexity is essential for computer scientists writing performance-critical software, as cache misses and memory access patterns directly interact with these physical limitations.

# REFERENCES

REF 01

## What Every Programmer Should Know About Memory

Author: Ulrich Drepper

REF 02

## TRM Zynq Ultrascale+: DDR Configuration

Source: AMD Documentation

REF 03

## DDR5 New DFE Features Explained

Source: Keysight Training